

LIAM SHAW

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EDUCATION

FLAME University, Pune · B.Sc. (Hons) Computer Science, Minor: Business Analytics · Full Scholarship 2022 – 2026
La Martiniere for Boys, Kolkata · ISC 95% (2022) · ICSE 92.4% (2020) · SAT 1540/1600 (99th percentile)

TECHNICAL SKILLS

ML / AI: PyTorch · HuggingFace Transformers · scikit-learn · XGBoost · CNNs · BiLSTMs · NLP · RAG · NL2SQL · PCA · Random Forest · KMeans · DistilBERT · RoBERTa · ResNet-18

Stack: Python · Java · JavaScript · TypeScript · SQL · FastAPI · Flask · Spring Boot · React · Docker · SQLite · PostgreSQL · HTML/CSS

Tools: CleverTap · pandas · FuzzyWuzzy · Git · Jira · Jupyter · Maven · Bubble.io

PROJECTS

MoodTrack — Multimodal Music Emotion Recognition · Graduation Project Mar 2026

- Segment-level arousal/valence tracking on 1,802 DEAM songs + 277 listener profiles (MERP); **late-fusion CNN-BiLSTM + DistilBERT** achieved arousal CCC 0.774, valence CCC 0.441; audio-only baseline 0.738.
- Personalised tri-branch model (5 listener archetypes); arousal CCC 0.533 — statistically distinct emotion trajectories by demographic. Learned gates autonomously replicated music-psychology valence-lyric finding.
- Deployed Flask web app: MP3 upload, Chart.js live trajectory visualisation, Euclidean persona-aware recommendation engine. Stack: PyTorch · librosa · HuggingFace · Flask.

ScoutAI — GenAI Football Scouting Platform · Applied Generative AI Apr 2026

- Full-stack AI app: FastAPI + SQLite (~370 MB) + JWT auth; **NL2SQL query engine**, chain-of-thought automated scouting reports, agentic transfer recommendation engine.
- ML subsystem: PCA, Random Forest, KMeans archetypes, cosine-similarity player search, RAG-enhanced LLM inference (Groq llama-3.1-8b + Ollama fallback); 20+ REST endpoints.

Hybrid Fake News Detection (AMR + RoBERTa) · NLP Research Dec 2025

- 87.07% accuracy** on FakeNewsNet — vs. 60–70% ceiling for standard baselines — combining RoBERTa embeddings with 26 hand-engineered AMR structural features (entity, syntactic, semantic, sentiment).
- Threshold optimisation (0.5 → 0.1) added ~3% accuracy; ablation confirmed AMR features as primary differentiator over pure transformer baseline.

FLAME Moodle Insight Dashboard · Software Architecture (CSIT306) · Team Project 2025

- React + TypeScript** analytics dashboard for FLAME's Center for Digital Learning — replacing a manual, hours-long Moodle CSV-to-Excel student engagement workflow with a local-first, privacy-preserving dashboard.
- Weighted scoring + traffic-light flagging surfaces at-risk students automatically; 4 client consultation rounds with the CDL team; deployed and in active use.

EXPERIENCE

ACEplus (Derek O'Brien & Associates) · Analytics & CRM Intern · Kolkata May – Jul 2025

- Designed **Zone Module Action** naming convention; audited 50+ inconsistent CleverTap event keys; authored 50+ event spec + 3-phase migration plan — formally adopted by the engineering team.
- Cohort analysis: workshop-driven users earned **~2x more ACEs** vs. organic signups — finding adopted by product team to prioritise school engagement programmes.
- Python + FuzzyWuzzy deduplication resolved ~15% of inconsistent user profiles; built user segments and audience filters in CleverTap; configured push, in-app, and email automation campaigns.

Mandrake Tech · Software Intern · Pune Oct 2024 – Jan 2025

- Built a containerised URL-shortening service (**Spring Boot, PostgreSQL, JWT auth**); Docker + Maven configuration removed environment-parity issues across the dev team.
- Implemented REST-integrated frontend interfaces; owned the full request lifecycle from UI through API layer to database persistence.

Willwali · Software Developer Intern · Remote May – Jul 2024

- Flask/Jinja CRUD microservices; **50+ unit and end-to-end tests at 90%+ coverage**; found and documented 30+ bugs in the Bubble will-creation MVP.
- Validated automated will outputs against Hindu Succession Act, Islamic Faraid, and Malaysian inheritance law edge cases.

CERTIFICATIONS

CS50x: Introduction to Computer Science (HarvardX) · **Building AI Literacy** (LinkedIn Learning) · **Responsive Web Design** (freeCodeCamp)